

Experimental Methods in Computer Science
(Metodologias Experimentais em Informática)

Henrique Madeira

Teaching assistants: ChatGPT, Claude, Gemini, Copilot, Perplexity, and alike

Master in Informatics Engineering
Departamento de Engenharia Informática
Faculdade de Ciências e Tecnologia da Universidade de Coimbra
2025/2026

Henrique Madeira, DEI-FCTUC, 2018-2026

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Course profile

**Experimental Methods in Computer Science
(and Informatics Engineering)** ←

- **Scientific Area:** Informatics
- **School Year:** 1st year MEI
- **Semester:** 1st semester
- **Classes**

- Lectures (T), concepts and discussion: 2:00h per week
- Practice-Lab. (PL), training exercises: 2:00h per week

- **ECTS:** 6
- **Teacher: Henrique Madeira**
 - Email: henrique@dei.uc.pt
 - Office: D3.2
 - Available for students: Wednesday, from 9:00h to 14:00h. And at any time through email or in the virtual room mei-2026

Goal: 30 to 40 minutes lectures followed by dedicated time for questions and discussions as necessary, with the aim of fostering greater student autonomy in the learning process.

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Teacher profile

Henrique Madeira

- **SSE Group** (Software and Systems Engineering) from **CISUC** (Centre for Informatics and Systems of University of Coimbra)
- Works on computer dependability, software engineering, and AI in safety critical systems
- Co-founder of Critical Software SA
- Some recent research projects:
 - **SATERA** (Space-based composite Ads-b and multilateration system validation through scalable simulations), H2020-SESAR, 2024-2026
 - **ADVANCE** (Addressing Verification and Validation Challenges in Future Cyber-Physical Systems), H2020, 2018-2025
 - **BUBBLES** (Defining the Building Basic Blocks for a U-Space Separation Management Service), H2020-SESAR, 2020-2023
 - **VALU3S** (Verification and Validation of Automated Systems' Safety and Security), H2020-ECSEL and FCT, 2020-2023
 - **AI4EU** (A European AI On Demand Platform and Ecosystem), H2020, 2018-2022
 - **BASE** (Biofeedback Augmented Software Engineering), FCT, 2018-2024

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Syllabus

- The experimental method: experimental scenarios in computer science and informatics engineering
- Experiment design (covering different types)
- Exploratory data analysis
- Data analysis: distributions, correlation, linear regression, etc.
- Measurements and benchmarking
- Hypothesis testing
- Experiments with people
- Experimental software engineering: overview and examples of experiments
(do not follow exactly this order in the lectures)

**We need basic statistics
But that is easy...**

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Assignment

- **Learning objectives:** design, implement, run, collect raw data, analyse the data, test hypotheses, and report the results.
- **Goal:** evaluate and compare different LLM based tools in the following tasks:
 - Code generation
 - Code explanation
 - Code review (bug discovery)
- Deadline: **May 18, 2026**
- The final output of the assignment include:
 - Written report.
 - Reproducibility package
- The assignment is a group activity:
 - Up to 4 students per group
 - **Evaluation is individual**

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Course organization

- Lectures (T):
 - Overview of the topics of the syllabuses
 - Discussion of key aspects
 - Discussion of examples
- Practice-Lab. (PL):
 - Exercises for concept consolidation and preparation for the written tests/exams.
 - The goal is to learn in the PL classes
 - At least 70% of attendance is required to be eligible for the written tests

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Course planning: lectures (T classes)

Updated version 1.4, 18 March

Experimental Methods in Computer Science, master plan

Experimental Methods in Computer Science - Plan for the T classes	
Dates	Activities
February 11, 2026	Introduction of the course
February 18, 2026	Design of experiments and basic aspects of hypothesis testing
February 25, 2026	Measurements in computer systems and software with confidence intervals
March 4, 2026	Hypothesis testing: methods for different scenarios and data distributions: test for a mean value
March 11, 2026	Hypothesis testing: methods for different scenarios and data distributions: test for a mean value
March 18, 2026	Hypothesis testing: methods for different scenarios and data distributions: more test for a mean value
March 25, 2026	Hypothesis testing: methods for different scenarios and data distributions: test for a proportion
April 8, 2026	Hypothesis testing: dependent samples + Written test 1 (Frequência 1)
April 15, 2026	Non-parametric statistical inference: Kolmogorov-Smirnov
April 22, 2026	Non-parametric statistical inference: Mann-Whitney U and other non-parametric methods
April 29, 2026	Hypothesis testing: one way ANOVA
May 6, 2026	Hypothesis testing: two way ANOVA
May 13, 2026	Written test 2 (Frequência 2)
May 22, 2026	Assignment defences

This is a tentative plan: most likely, we will need to adjust it during the semester.

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Course planning: PL classes

Experimental Methods in Computer Science - Plan for the PL classes	
Dates	Activities
February 11/12, 2026	No classes.
February 18/19, 2026	No classes
February 25/26, 2026	Training exercises and discussion: experiment design and measurements
March 04/05, 2026	Training exercises and discussion: hypothesis testing basics
March 11/12, 2026	Training exercises and discussion: hypothesis testing for different scenarios
March 18/19, 2026	Training exercises and discussion: more on hypothesis testing
March 25/26, 2026	Training exercises and discussion: test for a proportion
April 08/09, 2026	Assignment follow up and support
April 15/16, 2026	Training exercises and discussion: non-parametric statistical inference
April 22/23, 2026	Training exercises and discussion: more on non-parametric methods
April 29/30, 2026	Training exercises and discussion: dependent samples
May 06/07, 2026	Training exercises and discussion: ANOVA (one way and two way)
May 13/14, 2026	Assignment follow up and support (assignment deadline: May 18)
May 20/21, 2026	Assignment defences (classes times + extra time slots)

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Bibliography

- Documents provided by the teacher and papers available online
- “Experimental Methods for the Analysis of Optimization Algorithms”, T. Bartz-Beielstein, M. Chiarandini, L. Paquete, M. Preuss, Springer, 2010
- “Basics of Software Engineering Experimentation”, Natalia Juristo and Ana M. Moreno, Springer Publishing Company, 2010.
- NIST/SEMATECH e-Handbook of Statistical Methods:
<https://itl.nist.gov/div898/handbook/>
- “The Art of Computer Systems Performance Analysis”, R. Jain, Wiley 1991.
- “Measuring Computer Performance”, D. J. Lilja, Cambridge University Press, 2000

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Tools and other resources

For the assignments and for the PL classes

- R (<http://www.r-project.org/>)
- MS Excel, SPSS, or any other statistical application
- Online statistics calculator tools
 - <https://www.socscistatistics.com/tests/>
 - <https://www.meta-calculator.com/statistics-calculator.php>
 - (there are many more!)
- Statistical tables (for the written exams)

No problem in using LLM tools such as ChatGPT, provided that these tools are used to learn the concepts and skills defined as learning targets for this course.

Obviously, students' knowledge and skills will be evaluated without such tools.

"A fool with a tool is still a fool."

It is not clear who is the author of this quote... Grady Booch?, Kishor Trivedi?, ??

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Assessment

- Written tests: **65%**
 - Written test 1 – 50%
8 April 2026, at 9h00 (rooms will be assigned later)
 - Written test 2 – 50%
13 May 2026, at 9h00 (rooms will be assigned later)
 - Students may use two handwritten A4 sheets written in blue pen (*“teste sem consulta livre”*)
 - Requires at least 70% attendance of PL classes to be eligible for each written test
 - Minimum of 35% in each written test
- Written exams (normal and extra season/”recurso ”): **65%**
 - Dates and rooms will be defined by the Department
 - Questions cover all the topics addressed in the classes
 - Students may use two handwritten A4 sheets written in blue pen (*“exame sem consulta livre”*)
- Assignment + defence with individual questions: **35%**

Two attempts to succeed:

- Written tests/exam
- “Recurso ”

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Other issues

- Classes are highly relevant to save time for students...
- Plagiarism → Mandatory fail in the course + internal (UC) disciplinary procedure.
- Questions and difficulties → Henrique Madeira <henrique@dei.uc.pt>
- Feedback from students is very welcome ☺

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Questions?
Suggestions?

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